# Package ‘manMetaVAR’

February 24, 2026

**Title** Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients

**Version** 0.9.7

**Description** Research compendium for the manuscript
Pesigan, I. J. A., et al. (In Preparation).
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**URL** https://github.com/jeksterslab/manMetaVAR,
https://jeksterslab.github.io/manMetaVAR/,
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**BugReports** https://github.com/jeksterslab/manMetaVAR/issues

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# Contents



---

`Compress` *Compress Replication*

---

## Description

Compress Replication

## Usage

```
Compress(taskid, repid, output_folder)
```

## Arguments

| | |
|---|---|
| `taskid` | Positive integer. Task ID. |
| `repid` | Positive integer. Replication ID. |
| `output_folder` | Character string. Output folder. |

## Value

The output is saved as an external file in `output_folder`.

## Author(s)

Ivan Jacob Agaloos Pesigan

`FitDTVAR` *Fit the Model using the fitVARMxID Package*

## Description

The function fits the model using the fitVARMxID::fitVARMxID package.

## Usage

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

## Arguments

| | |
|---|---|
| `data` | R object. Output of the `GenData()` function. |
| `ncores` | Positive integer. Number of cores to use. |
| `seed` | Integer. Random seed. |

## See Also

Other Model Fitting Functions: `FitMetaVAR()`, `FitMplus()`, `MplusInput()`

## Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

`FitMetaVAR` *Multivariate Meta-Analysis using the metaDyn Package*

## Description

The function performs multivariate meta-snalysis using the metaDyn::metaDyn package.

## Usage

```
FitMetaVAR(fit, ncores = NULL, seed = NULL)
```

## Arguments

| | |
|---|---|
| `fit` | R object. Output of the `FitDTVAR()` function. |
| `ncores` | Positive integer. Number of cores to use. |
| `seed` | Integer. Random seed. |

## See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMplus()`, `MplusInput()`

## Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
meta <- FitMetaVAR(fit = fit, seed = seed)
summary(meta)
print(meta)
coef(meta)
vcov(meta)

## End(Not run)
```

---

`FitMplus` *Fit the Model using Mplus*

---

## Description

The function fits the model using Mplus.

## Usage

```
FitMplus(
  data,
  chains = 2L,
  iter = 20000L,
  fscores = NULL,
  plot = FALSE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL,
  seed = NULL
)
```

## Arguments

| | |
|---|---|
| `data` | R object. Output of the `GenData()` function. |
| `chains` | Integer. Number of chains. |
| `iter` | Integer. Number of iterations. |
| `fscores` | Integer. Number of iterations for factor scores. |
| `plot` | Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file. |
| `wd` | Character string. Working directory. |
| `mplus_bin` | Character string. Path to Mplus binary. If `mplus_bin = NULL`, the function will try to find the appropriate binary. |
| `ncores` | Positive integer. Number of cores to use. |
| `seed` | Integer. Random seed. |

## See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMetaVAR()`, `MplusInput()`

## Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

---

# `GenData` *Simulate Data*

---

## Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

## Usage

```
GenData(taskid, seed = NULL)
```

## Arguments

| | |
|---|---|
| `taskid` | Positive integer. Task ID. |
| `seed` | Integer. Random seed. |

## Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
print(data)
summary(data)
plot(data)

## End(Not run)
```

---

manmetavar-data-methods

*Methods for Objects of Class* `manmetavar.data`

---

## Description

This page documents the available methods for objects of class `manmetavar.data`.

## Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.data'
plot(x, ...)
```

## Arguments

| | |
|---|---|
| `x` | Object of class `manmetavar.data`. |
| `...` | additional arguments. |
| `object` | Object of class `manmetavar.data`. |

## Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-dtvar-methods
*Methods for Objects of Class* `manmetavar.dtvar`

## Description

This page documents the available methods for objects of class `manmetavar.dtvar`.

## Usage

```
## S3 method for class 'manmetavar.dtvar'
coef(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
vcov(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  robust = FALSE,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
```

```
  digits = 4,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)
```

## Arguments

| | |
|---|---|
| `object` | Object of class `manmetavar.dtvar`. |
| `mu` | Logical. If `mu` = `TRUE`, include estimates of the `mu` vector, if available. If `mu` = `FALSE`, exclude estimates of the `mu` vector. |
| `alpha` | Logical. If `alpha` = `TRUE`, include estimates of the `alpha` vector, if available. If `alpha` = `FALSE`, exclude estimates of the `alpha` vector. |
| `beta` | Logical. If `beta` = `TRUE`, include estimates of the `beta` matrix, if available. If `beta` = `FALSE`, exclude estimates of the `beta` matrix. |
| `nu` | Logical. If `nu` = `TRUE`, include estimates of the `nu` vector, if available. If `nu` = `FALSE`, exclude estimates of the `nu` vector. |
| `psi` | Logical. If `psi` = `TRUE`, include estimates of the `psi` matrix, if available. If `psi` = `FALSE`, exclude estimates of the `psi` matrix. |
| `theta` | Logical. If `theta` = `TRUE`, include estimates of the `theta` matrix, if available. If `theta` = `FALSE`, exclude estimates of the `theta` matrix. |
| `ncores` | Positive integer. Number of cores to use. |
| `...` | additional arguments. |
| `robust` | Logical. If `TRUE`, use robust (sandwich) sampling variance-covariance matrix. If `FALSE`, use normal theory sampling variance-covariance matrix. |
| `x` | Object of class `manmetavar.dtvar`. |
| `means` | Logical. If `means` = `TRUE`, return means. Otherwise, the function returns raw estimates. |
| `digits` | Integer indicating the number of decimal places to display. |

## Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-metavar-methods
*Methods for Objects of Class* manmetavar.metavar

---

## Description

This page documents the available methods for objects of class manmetavar.metavar.

## Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.metavar'
vcov(object, robust = NULL, ...)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, robust = NULL, ...)
```

## Arguments

| | |
|---|---|
| object | Object of class manmetavar.metavar. |
| ... | additional arguments. |
| robust | Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix. If NULL, the function will check object if robust standard errors are available. |
| x | Object of class manmetavar.metavar. |
| alpha | Numeric vector. Significance level $\alpha$. |
| digits | Integer indicating the number of decimal places to display. |
| parm | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level | the confidence level required. |

## Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-mplus-methods

*Methods for Objects of Class* manmetavar.mplus

## Description

This page documents the available methods for objects of class manmetavar.mplus.

## Usage

```
## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,
  burnin = NULL,
  legend_loc = "topright",
  ...
)
```

## Arguments

| | |
|---|---|
| object | Object of class manmetavar.mplus. |
| median | Logical. If median = TRUE, return median of the posterior. If median = FALSE, return mean of the posterior. |
| burnin | Integer indicating initial samples to discard. If burnin = NULL, do not discard anything. |
| ... | additional arguments. |
| alpha | Numeric vector. Significance level $\alpha$. |
| digits | Integer indicating the number of decimal places to display. |
| parm | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |

| | |
|---|---|
| `level` | the confidence level required. |
| `x` | Object of class `manmetavar.mplus`. |
| `what` | Character string. If `what = "posterior"`, return posterior distribution plots. If `what = "trace"`, return trace plots. |
| `legend_loc` | Character string. Legend location. |

### Author(s)

Ivan Jacob Agaloos Pesigan

---

`model` *Model Parameters*

---

### Description

Model Parameters

### Usage

```
data(model)
```

### Format

A list with 15 elements:

**k** Number of variables.

**mu_mu** Mean of the set-point vector $\boldsymbol{\mu}$.

**mu_sigma** Covariance matrix of the parameter $\boldsymbol{\mu}$.

**mu_sigma_l** Cholesky factor of the covariance matrix of the parameter $\boldsymbol{\mu}$.

**beta_mu** Mean of lagged coefficients matrix $\boldsymbol{\beta}$.

**beta_sigma** Covariance matrix of the parameter $\mathrm{vec}\left(\boldsymbol{\beta}\right)$.

**beta_sigma_l** Cholesky factor of the covariance matrix of the parameter $\mathrm{vec}\left(\boldsymbol{\beta}\right)$.

**psi** Process noise covariance matrix $\boldsymbol{\Psi}$.

**psi_l** Cholesky factor of the process noise covariance matrix $\boldsymbol{\Psi}$.

**psi_d_ldl** `uc_d` of the LDL' decomposition of the process noise covariance matrix $\boldsymbol{\Psi}$. See `fitVARMxID::LDL()`.

**psi_l_ldl** `s_l` of the LDL' decomposition of the process noise covariance matrix $\boldsymbol{\Psi}$. See `fitVARMxID::LDL()`.

**ma_fixed** Vector of fixed effects $\boldsymbol{\theta} = \left[\boldsymbol{\mu}, \mathrm{vec}\left(\boldsymbol{\beta}\right)\right]'$

**ma_random** Matrix of random effects $\boldsymbol{\theta} = \left[\boldsymbol{\mu}, \mathrm{vec}\left(\boldsymbol{\beta}\right)\right]'$

**ma_random_d_ldl** `uc_d` of the LDL' decomposition of the random effects. See `fitVARMxID::LDL()`.

**ma_random_l_ldl** `s_l` of the LDL' decomposition of the random effects. See `fitVARMxID::LDL()`.

### Author(s)

Ivan Jacob Agaloos Pesigan

MplusInput *Create Mplus Input File*

## Description

The function creates an Mplus input file.

## Usage

```
MplusInput(
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  ncores = NULL,
  seed = NULL
)
```

## Arguments

| | |
|---|---|
| `fn_data` | Character string. Filename for data file. |
| `fn_posterior` | Character string. Filename for posterior output. |
| `fn_factorscores` | Character string. Filename for factor scores output. |
| `chains` | Integer. Number of chains. |
| `iter` | Integer. Number of iterations. |
| `fscores` | Integer. Number of iterations for factor scores. |
| `plot` | Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file. |
| `ncores` | Positive integer. Number of cores to use. |
| `seed` | Integer. Random seed. |

## See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMetaVAR()`, `FitMplus()`

## Examples

```
cat(
  MplusInput(
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
```

```
      chains = 2L,
      iter = 20000L,
      fscores = 1000L,
      plot = TRUE,
      ncores = NULL,
      seed = 42
    )
  )
```

---

params *Simulation Parameters*

---

## Description

Simulation Parameters

## Usage

```
data(params)
```

## Format

A dataframe with 9 rows and 3 columns:

**taskid** Simulation Task ID.

**n** Sample size.

**time** Number of measurement occassions.

## Author(s)

Ivan Jacob Agaloos Pesigan

---

Sim *Simulation Replication*

---

## Description

Simulation Replication

### Usage

```
Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  metavar,
  mplus,
  chains,
  iter,
  fscores,
  plot
)
```

### Arguments

| | |
|---|---|
| taskid | Positive integer. Task ID. |
| repid | Positive integer. Replication ID. |
| output_folder | Character string. Output folder. |
| overwrite | Logical. Overwrite existing output in output_folder. |
| integrity | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| seed | Integer. Random seed. |
| data | Logical. Simulate data. |
| metavar | Logical. If metavar = TRUE, fit the model using the metaDyn package. |
| mplus | Logical. If mplus = TRUE, fit the model using DSEM in Mplus. |
| chains | Integer. Number of chains. |
| iter | Integer. Number of iterations. |
| fscores | Integer. Number of iterations for factor scores. |
| plot | Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file. |

### Value

The output is saved as an external file in output_folder.

### Author(s)

Ivan Jacob Agaloos Pesigan

SimFitDTVAR *Simulation Replication - FitDTVAR*

## Description

Simulation Replication - FitDTVAR

## Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

## Arguments

| | |
|---|---|
| taskid | Positive integer. Task ID. |
| repid | Positive integer. Replication ID. |
| output_folder | Character string. Output folder. |
| seed | Integer. Random seed. |
| suffix | Character string. Output of `manCTMed:::.SimSuffix()`. |
| overwrite | Logical. Overwrite existing output in `output_folder`. |
| integrity | Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`. |

## Details

This function is executed via the `Sim` function.

## Value

The output is saved as an external file in `output_folder`.

## Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR *Simulation Replication - FitMetaVAR*

## Description

Simulation Replication - FitMetaVAR

## Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

## Arguments

| | |
|---|---|
| `taskid` | Positive integer. Task ID. |
| `repid` | Positive integer. Replication ID. |
| `output_folder` | Character string. Output folder. |
| `seed` | Integer. Random seed. |
| `suffix` | Character string. Output of `manCTMed:::.SimSuffix()`. |
| `overwrite` | Logical. Overwrite existing output in `output_folder`. |
| `integrity` | Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`. |

## Details

This function is executed via the `Sim` function.

## Value

The output is saved as an external file in `output_folder`.

## Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplus *Simulation Replication - FitMplus*

## Description

Simulation Replication - FitMplus

## Usage

```
SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot
)
```

## Arguments

| | |
|---|---|
| taskid | Positive integer. Task ID. |
| repid | Positive integer. Replication ID. |
| output_folder | Character string. Output folder. |
| seed | Integer. Random seed. |
| suffix | Character string. Output of `manCTMed:::.SimSuffix()`. |
| overwrite | Logical. Overwrite existing output in `output_folder`. |
| integrity | Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`. |
| chains | Integer. Number of chains. |
| iter | Integer. Number of iterations. |
| fscores | Integer. Number of iterations for factor scores. |
| plot | Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file. |

## Details

This function is executed via the `Sim` function.

## Value

The output is saved as an external file in `output_folder`.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

`SimFN` *Simulation File Name*

---

### Description

Simulation File Name

### Usage

```
SimFN(output_type, output_folder, suffix)
```

### Arguments

| | |
|---|---|
| `output_type` | Character string. Output type. Valid values include `"data"`, `"fit-dt-var-mx"`, `"fit-meta-var-mx"`, and `"fit-mplus"`. |
| `output_folder` | Character string. Output folder. |
| `suffix` | Character string. Output of `manCTMed:::.SimSuffix()`. |

### Value

Returns a character string file name with the `output_folder` in the OS-specific format.

---

`SimGenData` *Simulation Replication - GenData*

---

### Description

Simulation Replication - GenData

### Usage

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

### Arguments

| | |
|---|---|
| `taskid` | Positive integer. Task ID. |
| `repid` | Positive integer. Replication ID. |
| `output_folder` | Character string. Output folder. |
| `seed` | Integer. Random seed. |
| `suffix` | Character string. Output of `manCTMed:::.SimSuffix()`. |
| `overwrite` | Logical. Overwrite existing output in `output_folder`. |
| `integrity` | Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`. |

## Details

This function is executed via the `Sim` function.

## Value

The output is saved as an external file in `output_folder`.

## Author(s)

Ivan Jacob Agaloos Pesigan

---

`SimProj` *Simulation Project Name*

---

## Description

Simulation Project Name

## Usage

```
SimProj()
```

## Value

Returns the project name as a character string.

## Author(s)

Ivan Jacob Agaloos Pesigan

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